

DATA SOURCES

Raw Logs & Semantic Corpus

PARALLEL RETRIEVAL STREAMS

Domain-Specific Scoring Channels

PER-USER ALIGNMENT

Dynamic Range Normalization

MULTI-STREAM ARBITER

Global Weighted Convex Fusion

CANDIDATE SERVING

Masking, Exact Retrieval & Top-K

Last.fm 1K Events

Chronological Log Split

19,150,865 Events

992 Users | 1.49M Items

Train: $t < 2009-04-01$

Val: [2009-04, 2009-05)

Test: [2009-05, 2009-07)

Zero-Leakage Cutoff

Play Counts

C_u Weights

HetRec 2011 Tags

Artist-Level Metadata

2,299 Filtered Tags

594,225 Items Covered

39.78% Catalog Coverage

11.92M Matrix NNZ

Sparse CSR TF-IDF

Timestamps

Tag Matrix T

STREAM 1: Collaborative

Implicit MSVD (ALS)

$$c_{ui} = 1 + \kappa \ln(1 + r_{ui})$$

Latent Factors: $K = 64$

$$\lambda = 0.05, \kappa = 40.0$$

10 Full ALS Iterations

$$\text{Score: } s_{ui}^{\text{msvd}} = x_u^T y_i$$

s_{ui}^{msvd}

STREAM 2: Semantic

HetRec Tag TF-IDF

Vocabulary: $|\mathcal{V}| = 2,299$

$$\text{Track Vector: } t_i = \text{TF-IDF}(i) / \|\cdot\|_2$$

$$\text{User Profile: } z_u = C_u T / \|C_u T\|_2$$

$$\text{Score: } s_{ui}^{\text{tag}} = z_u^T t_i$$

s_{ui}^{tag}

STREAM 3: Temporal

Temporal Recency Preference

$$\Delta t_{ui} = (t_{\text{ref}} - t_{ui}) / 86,400$$

$$\text{Decay Rate: } \gamma = 0.10 \text{ day}^{-1}$$

$$s_{ui}^{\text{temp}} = c_{ui} \exp(-\gamma \Delta t_{ui})$$

$$\text{Score: } s_u^{\text{temp}} = s_u^{\text{temp}} / \|s_u^{\text{temp}}\|_2$$

s_{ui}^{temp}

Score Alignment Layer

Per-User Calibration

Collaborative Min-Max:

$$\tilde{s}_{ui}^{\text{msvd}} = \frac{s_{ui}^{\text{msvd}} - \min_j s_{uj}^{\text{msvd}}}{\max_j s_{uj}^{\text{msvd}} - \min_j s_{uj}^{\text{msvd}}}$$

Maps $(-\infty, \infty) \rightarrow [0, 1]$

Resolves user score variance

Auxiliary Stream Alignment:

Cosine Tag Score: $\tilde{s}_{ui}^{\text{tag}} \in [0, 1]$

L2 Temporal: $\tilde{s}_{ui}^{\text{temp}} \in [0, 1]$

Cross-Stream Scale Compatibility

Aligned Streams

Global Normalized Arbiter

Multi-Stream Fusion

Global Mixing Weights:

$$w_{\text{msvd}} = 0.20$$

$$w_{\text{tag}} = 0.30$$

$$w_{\text{temp}} = 0.50$$

Fused Scoring Function:

$$\hat{r}_{ui} = 0.20 \tilde{s}_{ui}^{\text{msvd}}$$

$$+ 0.30 \tilde{s}_{ui}^{\text{tag}}$$

$$+ 0.50 \tilde{s}_{ui}^{\text{temp}}$$

Validation Optimized

Weights strictly frozen

Score Matrix

Candidate Masking

Seen Interaction Filter

Historical Set: $\mathcal{H}_u = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}}$

Masking: $\hat{r}_{ui} \leftarrow -\infty \quad \forall i \in \mathcal{H}_u$

Pure Novel Discovery

Exact Top-K Retrieval

Full Catalog Ranking

Catalog: $|I| = 1,493,688$ Items

Exact numpy.argpartition

No ANN / Graph Approximations

Latency: 96.48 ms (10.36 qps)

Sparse Memory Safe (0 MB Dense)

Top-K Recommendations

Ordered Ranking & Attribution

$$\text{Top-}K(u) = \arg \text{top}K_i \hat{r}_{ui}$$

Evaluated at $K \in \{5, 10, 20\}$

NDCG@10: 0.007414

Low-Activity Gain: +50.47%

Linear Contribution Attribution:

$$\text{Contrib}_m(u, i) = \frac{w_m s_{ui}^m}{\hat{r}_{ui}} \times 100\%$$